# Global Strategic Narrative Monitor

## Mapping the changing strategic narratives of states through NLP

### Working product name

**Strategic Gravity**

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## 1. The idea in one sentence

The Global Strategic Narrative Monitor is an interactive NLP-based platform that shows how the official rhetoric of countries moves through global geopolitical narrative space over time, allowing users to see which strategic narratives countries are moving closer to or further away from and to inspect the actual speeches and statements driving those movements.

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# 2. The basic idea

Imagine the world as a strategic universe.

Certain countries or strategic narratives function like large gravitational centres.

For example, on the Russia–Ukraine issue, the important narrative centres might include:

- Russia
- Ukraine
- United States / NATO
- a negotiation or non-alignment narrative

Countries such as India, Brazil, South Africa, Hungary, Türkiye, Morocco, Indonesia or China would appear as other objects in this space.

Their position is not fixed.

A country might be relatively close to one narrative in 2021, move dramatically after February 2022, move again in 2023, and partially return toward another narrative by 2025.

A year slider allows the user to watch this movement.

Clicking a country does not simply say:

> “India = 62% Russia.”

Instead, it explains why the model placed India there.

The user would see:

**India — 2023**

Narrative proximity:

- Russian narrative: moderate
- Ukrainian narrative: low/moderate
- Western/NATO narrative: moderate
- non-alignment/negotiation narrative: high

Important themes:

- sovereignty
- diplomatic resolution
- energy security
- Global South
- strategic autonomy

Representative statements:

- actual excerpts from Indian official speeches

Change from 2022:

- stronger emphasis on negotiation
- continued reluctance to use some Western framing
- increased reference to sovereignty and territorial integrity

That last part — **showing the evidence** — is extremely important.

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# 3. What the project is actually measuring

The project is **not trying to determine which countries are friends or enemies**.

It is measuring something narrower:

> How similar or different are the ways governments publicly frame important geopolitical issues?

This distinction is very important.

Two countries can cooperate militarily while using quite different rhetoric.

Two countries can use similar language about “peace” while supporting completely different political outcomes.

A government can also change its rhetoric without changing its actual policy.

Therefore the core object you are studying is:

# Strategic narrative

In simple terms, a strategic narrative is:

> The story a government tells about what is happening in the world, why it is happening, who is responsible, what values or interests matter, and what should happen next.

For Ukraine, for example, different narratives might emphasize:

**Narrative A**

- aggression
- territorial integrity
- international law
- Russian responsibility
- military support for Ukraine

**Narrative B**

- NATO expansion
- Western responsibility
- Russian security
- multipolarity

**Narrative C**

- ceasefire
- negotiation
- avoiding escalation
- neutrality
- economic consequences for the Global South

Countries do not necessarily fit completely inside one narrative.

That is exactly why your “gravity” idea works.

They can sit **between narratives**.

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# 4. The main research question

The broad research question would be:

> **How do the strategic narratives of states converge, diverge and shift over time in response to major geopolitical developments?**

You can then have several smaller questions.

### Question 1 — Position

Where does each country's official rhetoric sit within the narrative space surrounding a geopolitical issue?

### Question 2 — Movement

How does that position change over time?

### Question 3 — Convergence

Which countries are becoming rhetorically more similar?

### Question 4 — Divergence

Which traditional partners are becoming rhetorically less similar?

### Question 5 — Turning points

When do major narrative changes happen?

### Question 6 — Explanation

Which specific statements, topics and frames explain these movements?

### Question 7 — Rhetoric versus behaviour

Do countries speak similarly but behave differently?

This last question could eventually become one of the project's most interesting features.

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# 5. Do NOT start with every country and every conflict

The final vision can be global.

The first version should not be.

Trying to scrape every president, foreign ministry, parliament and government in the world as your first step will probably kill the project.

Instead, build it in stages.

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# 6. Version 1 — the MVP

MVP means **Minimum Viable Product**.

It means:

> the smallest version of the project that already demonstrates that your idea works.

For Version 1, I recommend:

## Time period

**2014–2025**

This gives you:

- annexation of Crimea
- pre-2022 period
- 2022 invasion
- several years after the invasion
- enough years to visualize movement

## Countries

Start with approximately **25–30 countries**.

For example:

**Major powers**

- United States
- China
- Russia
- India

**Europe**

- Ukraine
- Netherlands
- France
- Germany
- Poland
- Hungary
- Italy
- United Kingdom

**Middle East**

- Türkiye
- Israel
- Iran
- Saudi Arabia
- UAE

**Africa**

- South Africa
- Morocco
- Kenya
- Nigeria
- Egypt

**Latin America**

- Brazil
- Mexico
- Argentina

**Asia-Pacific**

- Japan
- South Korea
- Indonesia
- Australia

The precise list can change.

The important point is that you include states from different geopolitical positions rather than only Europe and North America.

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# 7. Your first issue

Even though the platform is eventually global and multi-topic, I would still build the first prototype around:

## Russia–Ukraine and the international security order

Why?

Because it gives you:

- obvious geopolitical differences
- large amounts of text
- major changes over time
- observable events
- many countries adopting distinct positions
- a clear test of whether your visualization makes sense

Once this works, you can add other geopolitical spaces.

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# 8. Future issue modules

Eventually the user could choose an issue from a menu.

### Russia–Ukraine

Possible narratives around:

- sovereignty
- territorial integrity
- NATO
- sanctions
- negotiation
- security guarantees
- escalation

### US–China / Taiwan

Themes:

- One China
- sovereignty
- Indo-Pacific
- freedom of navigation
- strategic competition
- economic security
- peaceful reunification

## Israel–Palestine / Gaza

Themes:

- self-defence
- ceasefire
- humanitarian law
- Palestinian statehood
- terrorism
- occupation
- civilian protection

## Global order / multipolarity

Themes:

- rules-based international order
- UN-centred multilateralism
- Western dominance
- Global South
- BRICS
- sovereignty
- strategic autonomy

## NATO and European security

Themes:

- deterrence
- strategic autonomy
- transatlantic cooperation
- defence spending
- burden sharing
- Russia
- collective defence

The important methodological point is:

**Each geopolitical issue gets its own narrative space.**

There should not be one magical number saying:

> Brazil = 47% West, 53% Russia/China.

International politics is much more complicated than that.

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# 9. Your first dataset

This is where I would make your life much easier.

## Start with UN General Debate speeches

Every year, representatives of almost every country speak at the UN General Assembly.

These speeches are perfect for the first version because countries are speaking:

- in approximately the same institutional setting
- around the same period every year
- about international affairs
- using official government rhetoric

This makes countries much easier to compare.

There is already a **United Nations General Debate Corpus**, so you do not initially need to build a complicated web scraper.

The corpus contains thousands of speeches from countries around the world and covers decades.

Your basic dataset therefore looks like:

| country | year | speech |
|---|---|---|
| Netherlands | 2022 | full text... |
| India | 2022 | full text... |
| Brazil | 2022 | full text... |
| Russia | 2022 | full text... |
| Netherlands | 2023 | full text... |

This is a beautiful dataset for learning NLP.

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# 10. The first important NLP problem

Suppose India's UN speech is 4,000 words.

It might discuss:

- Ukraine
- climate change
- poverty
- terrorism
- UN reform
- development
- Palestine
- nuclear weapons

You cannot use the entire speech to calculate India's Ukraine position.

That would introduce noise.

So you need to break speeches into smaller pieces.

This is called:

# Chunking

Very simply:

> Chunking means cutting a large text into smaller pieces.

For example:

Speech → paragraphs.

Each paragraph becomes one observation.

Your data might become:

| country | year | paragraph | relevant_to_ukraine |
|---|---|---|---|
| India | 2023 | paragraph 1 | No |
| India | 2023 | paragraph 2 | No |
| India | 2023 | paragraph 3 | Yes |
| India | 2023 | paragraph 4 | Yes |
| India | 2023 | paragraph 5 | No |

Then you analyse only relevant paragraphs.

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# 11. How does the computer know what a paragraph means?

This is where you learn one of the most important concepts in modern NLP:

# Embeddings

The word sounds intimidating.

The concept is straightforward.

Imagine the sentence:

> Russia must withdraw its forces from Ukrainian territory.

A computer does not understand the sentence in the same way you do.

An embedding model converts that sentence into a long list of numbers.

Something like:

[0.12, -0.31, 0.84, 0.07, ...]

There may be hundreds of numbers.

You don't need to interpret individual numbers.

Together they represent something about the **meaning of the text**.

Texts with similar meanings tend to receive similar numerical representations.

For example:

> Russia should withdraw from Ukrainian territory.

and:

> Moscow must remove its forces from Ukraine.

should have relatively similar embeddings even though the exact words differ.

That is extremely useful for your project.

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# 12. Similarity

Once two texts have embeddings, you can calculate how similar they are.

The common measure is:

## Cosine similarity

You do not need to be scared by the mathematics.

Conceptually:

- 1 = very similar
- around 0 = unrelated
- lower values = very different

For your project, you might compare an Indian statement with a collection of Ukrainian statements.

And then with Russian statements.

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# 13. But similarity alone is dangerous

Suppose France says:

> Russia's description of NATO expansion is entirely false.

A basic semantic model might notice:

- Russia
- NATO expansion
- security

and decide that the French paragraph resembles Russian discourse.

But France is actually **rejecting** that narrative.

Therefore you need another concept:

# Stance detection

Stance means:

> What position is the speaker taking toward an idea?

For example:

Sentence:

> Sanctions against Russia must be strengthened.

Stance toward sanctions:

**Support**

Sentence:

> Western sanctions are damaging the global economy and should end.

Stance toward sanctions:

**Oppose**

Sentence:

> Several governments discussed the effects of sanctions.

Stance:

**Neutral / unclear**

This is much more useful than ordinary sentiment analysis.

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# 14. Sentiment is NOT enough

You may have previously used positive/negative sentiment analysis.

I would not make sentiment the main method here.

Consider:

> Ukraine's heroic resistance must continue.

Positive sentiment.

Versus:

> Russia's brutal invasion must end.

Negative sentiment.

Both represent broadly similar geopolitical positioning.

Therefore:

**positive/negative emotion is not the thing you care about.**

You care about:

- framing
- position
- concepts
- policy preferences
- responsibility
- proposed solutions

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# 15. Narrative dimensions

Before doing fancy machine learning, I would define several dimensions that are politically meaningful.

For Russia–Ukraine, for example:

### Responsibility

Who is presented as responsible for the conflict?

### Territorial integrity

How strongly is Ukrainian sovereignty emphasized?

### NATO

Is NATO presented as:

- defensive
- threatening
- irrelevant?

### Sanctions

Does the government:

- support them
- oppose them

- avoid discussing them?

### Military assistance

Support / opposition / neutrality.

### Negotiation

How strongly does the government emphasize diplomacy and ceasefire?

### Escalation

Does it emphasize dangers of escalation?

### International law

How central is legal framing?

### Global South effects

Food, energy, inflation, development consequences.

This creates an interpretable layer underneath the visualization.

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# 16. Narrative poles

Now we return to your gravity metaphor.

Instead of treating Russia itself as literally one giant black hole, we define:

## Narrative poles

A narrative pole is a collection of texts representing a recognisable geopolitical framing.

For example:

### Ukrainian narrative pole

Constructed from Ukrainian official statements concerning the conflict.

### Russian narrative pole

Constructed from Russian official statements.

### Western security narrative pole

Potentially constructed from US/NATO or selected allied statements.

### Negotiation/non-alignment pole

Potentially derived from countries strongly emphasizing ceasefire, dialogue and strategic autonomy.

Then every country's rhetoric can be compared to those poles.

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# 17. Important methodological improvement

Do not permanently define:

**Russia = Russian pole**

across every topic and every year.

Russia's own rhetoric changes.

So your anchors should ideally be:

**issue-specific and time-specific.**

For example:

Russia–Ukraine:

- Russia 2022 narrative
- Ukraine 2022 narrative

China–Taiwan:

- Chinese narrative
- Taiwanese narrative
- US Indo-Pacific narrative

That is much more defensible.

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# 18. Turning similarity into “gravity”

Suppose for Brazil in 2023 your model estimates:

- similarity with Ukraine narrative = 0.63
- similarity with Russia narrative = 0.48
- similarity with negotiation narrative = 0.78

You can transform these into relative gravitational weights.

Conceptually:

> The closer Brazil's discourse is to a narrative, the stronger that narrative's visual “pull.”

You do not have to initially create real physics equations.

The animation is a **metaphor for statistical similarity**.

In the methodology section of the project you explicitly explain this.

You might call the final quantity:

## Narrative Proximity Score

rather than:

**Political Alignment Score**.

That wording matters.

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# 19. Country-year-topic

This should become one of the central ideas of the project.

Your unit of analysis is not simply:

**country**

It is:

> **country × year × issue**

For example:

- India × 2022 × Ukraine
- India × 2023 × Ukraine
- India × 2024 × Ukraine

or:

- Brazil × 2024 × global order
- Brazil × 2024 × Gaza
- Brazil × 2024 × Ukraine

This allows the same country to occupy very different positions depending on the issue.

Which is exactly what happens in real diplomacy.

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# 20. Creating one representation for a country

Imagine that India has eight paragraphs about Ukraine in its 2024 speech.

Each paragraph gets an embedding.

You then combine them.

The simplest approach is to take their average.

This gives you something like:

## India–2024–Ukraine vector

That vector represents India's overall Ukraine-related discourse for that year.

You do the same for every country.

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# 21. The visualisation

Your main screen should be the memorable part of the project.

## Strategic Gravity view

Imagine:

**Ukraine** — large fixed object

**Russia** — large fixed object

**Negotiation/non-alignment narrative** — another centre

Around them:

🇫🇷 France
🇮🇳 India
🇧🇷 Brazil
🇿🇦 South Africa
🇭🇺 Hungary
🇲🇦 Morocco
🇩🇪 Germany
🇨🇳 China

etc.

The user moves:

**2018 → 2019 → 2020 → 2021 → 2022 → 2023 → 2024 → 2025**

Countries move.

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# 22. Trails

Do not only show the current position.

Show a faint trail.

For example:

India:

2021 ●———2022 ●———2023 ●———2024 ●

That immediately communicates:

**direction of narrative movement.**

You could calculate:

## Narrative displacement

Simply:

> How far did the country's narrative move compared with the previous year?

A very large movement becomes interesting.

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# 23. Narrative shocks

You could highlight unusually large movements.

For example:

### Narrative Shift Detected

Germany — 2022

Then the user can click it.

The system displays:

**Major change compared with 2021**

Potentially associated themes:

- Zeitenwende
- military assistance
- Russian aggression
- European security
- energy dependency

Supporting statements appear underneath.

This introduces something very relevant to strategic analysis:

## Change-point detection

A change point means:

> A moment where the pattern changes unusually strongly.

You can learn the actual statistical methods later.

For Version 1, simply calculating large year-to-year differences is sufficient.

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# 24. The evidence panel

This is possibly the most important feature.

Every point must be auditable.

The user clicks:

**🇧🇷 Brazil — 2023**

Then sees:

### Narrative position

Negotiation narrative — high proximity
Ukraine narrative — moderate proximity
Russia narrative — moderate proximity

### Main frames

1. negotiated settlement
2. food security
3. multipolarity
4. territorial integrity
5. Global South

### Representative evidence

Then show actual paragraphs from Brazilian statements.

This ensures that the system does not become:

> “The AI says Brazil is here. Trust us.”

The analyst can inspect the evidence.

That is a major strength.

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# 25. How do you choose the representative sentences?

Another useful NLP concept:

## Retrieval

Retrieval simply means:

> Find the pieces of text most relevant to something.

If Brazil is shown as moving toward a negotiation narrative, you search Brazil's relevant paragraphs for the ones most associated with that dimension.

Those are displayed.

This gives your system explainability.

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# 26. A second visualisation: the real narrative map

I would actually build **two visual modes**.

## Mode 1 — Strategic Gravity

Your intuitive visual.

The narrative poles occupy fixed positions.

Countries move between them.

This is designed for communication.

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## Mode 2 — Semantic Map

This is the more technical view.

Here you let the NLP model determine which documents sit close to each other.

For this you can eventually learn:

## UMAP

Again, intimidating name; easy concept.

Your embeddings may contain hundreds of dimensions.

Humans cannot visualize 384 dimensions.

UMAP takes those hundreds of dimensions and compresses them into:

- X

- Y

so you can draw them on a screen.

Important:

The X and Y axes do **not** mean something like left/right politics.

The important information is:

> points that appear close together have relatively similar language.

This gives your project both:

**intuitive visualization** + **methodological visualization**.

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# 27. Topics

Another NLP technique you will eventually learn is:

## Topic modelling

Topic modelling tries to automatically discover groups of words or documents that repeatedly occur together.

For example, the machine might discover a cluster containing:

- gas
- energy
- supply
- prices
- pipeline
- dependence

You interpret this as:

**Energy security**

Another might contain:

- territorial
- sovereignty
- charter
- borders
- aggression

You might interpret that as:

**International law / territorial integrity**

Modern approaches such as BERTopic use embeddings to help discover these clusters.

Do not start with BERTopic on Day 1.

It can come after your basic embedding pipeline works.

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# 28. Statistics: where your master's becomes valuable

You should not make this purely an NLP project.

Your statistical training is one of your advantages.

Eventually you can add several statistical components.

### Confidence / uncertainty

Instead of reporting:

> France's Ukraine proximity = 0.71

as if it is perfectly certain, estimate how stable that measurement is.

For example, calculate the score using different paragraphs.

If almost every paragraph gives a similar conclusion, confidence is high.

If France has only one relevant sentence, confidence should be low.

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### Trend estimation

Instead of only displaying individual years, estimate whether a country is generally moving toward or away from a narrative.

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### Change-point detection

Identify statistically unusual changes.

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### Clustering

Automatically find countries with similar narrative profiles.

Perhaps your model discovers clusters like:

- strong Western security framing
- negotiation-focused states
- sovereignty/non-intervention states
- Russian-aligned framing

You then investigate them qualitatively.

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# 29. The really interesting extension: rhetoric versus action

Once the narrative system works, add:

# Narrative–Behaviour Gap

This would ask:

> Does a country's rhetoric correspond to what the country actually does?

For example, use UN General Assembly voting.

Suppose two countries sound similar in speeches.

But then they repeatedly vote differently on Ukraine-related resolutions.

That is interesting.

Or the opposite:

They vote similarly but explain their positions using very different narratives.

You could create:

**Rhetorical proximity**

versus

**Behavioural proximity**

Then flag cases where the two disagree.

That becomes a genuine political-science question.

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# 30. Example

Imagine:

Country A publicly emphasizes:

- peace
- sovereignty
- negotiation
- neutrality

Its rhetoric therefore appears close to a non-alignment narrative.

But its UN voting behaviour consistently matches Western countries.

Your application could show:

### Narrative–Behaviour Gap: High

That is much more interesting than simply classifying the country.

An analyst would then ask:

> Why?

And that is where your international-relations background becomes useful.

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# 31. Data sources for the later validation layer

You could eventually combine your speech data with:

### UN General Assembly votes

Excellent because they are standardized and country-specific.

### Sanctions participation

Did the country impose sanctions?

### Military assistance

Where reliable comparable data exist.

### Trade

For example, changing economic relationships.

### Diplomatic visits

Potentially later.

### Conflict/event datasets

Such as ACLED or other structured event sources where appropriate.

But do not add all these immediately.

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# 32. The best data strategy

## Phase 1

UN speeches only.

## Phase 2

UN speeches + UN voting.

## Phase 3

Additional official government sources.

## Phase 4

News/event data.

This keeps the project manageable.

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# 33. Why UN speeches are such a strong starting point

You avoid several problems.

If you immediately scrape governments:

France might publish 5,000 statements.

Morocco might publish 400.

China might publish material in different languages.

Some websites are easy to scrape.

Others are terrible.

Some presidents speak constantly.

Others rarely do.

Comparability becomes difficult.

At the UN General Debate, governments are operating in approximately the same setting.

So you get a much cleaner starting dataset.

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# 34. Phase 2 data expansion

Later you can add:

### Foreign ministries

### Presidential speeches

### Prime-minister speeches

### Parliamentary debates

### UN Security Council interventions

### NATO speeches

### EU statements

Then the tool becomes more frequent than yearly.

Instead of:

**country × year**

you may eventually reach:

**country × month**

That would make the animation much more powerful.

But that is Version 2 or 3.

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# 35. Data collection skills you need to learn

You do not need to become an expert in everything.

Learn these in order.

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## Skill 1 — Python fundamentals

You need:

- variables
- functions
- loops
- lists
- dictionaries
- importing packages
- reading files

You probably already have some of this.

Do not spend three weeks doing generic Python courses.

Learn by working on this project.

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## Skill 2 — Pandas

Pandas handles tables in Python.

Think of it as:

**Excel for Python, but much more powerful.**

You should learn:

- read_csv()
- filtering rows
- selecting columns
- groupby()
- merge()
- apply()
- handling missing values

This will probably be used every day.

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## Skill 3 — Text preprocessing

Learn how to:

- normalize whitespace
- remove unnecessary characters
- split text into paragraphs
- identify sentences
- store text and metadata

Do not obsess over removing stopwords etc. immediately.

Modern transformer models usually need much less traditional cleaning than older NLP approaches.

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# 36. Skill 4 — NLP basics

Learn these concepts:

### Token

A small piece of text a language model processes.

### Corpus

A collection of documents.

Your collection of UN speeches is a corpus.

### Document

One individual piece of text.

### Embedding

Numbers representing the semantic meaning of text.

### Similarity

A numerical measure of how similar two embeddings are.

### Classification

Assigning a predefined category.

Example:

Ukraine-related / not Ukraine-related.

### Clustering

Finding groups without giving the categories beforehand.

### Topic modelling

Discovering recurring themes.

### Stance detection

Determining the position taken toward a claim or issue.

If you understand those concepts, you already understand much of what your system is doing.

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# 37. Skill 5 — Sentence Transformers

This will probably be your first important modern NLP library.

Python package:

**sentence-transformers**

It lets you do something like:

text
↓
embedding model
↓
vector

You can then compare sentences or paragraphs.

For your first prototype you do **not train your own model**.

Use a model somebody else has already trained.

That is called:

## Pretrained model

This saves enormous amounts of work.

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# 38. Skill 6 — Cosine similarity

Learn what it means and how to calculate it using Python.

You do not need advanced linear algebra.

You mainly need to understand:

> It compares the direction of two numerical vectors and gives us an indication of semantic similarity.

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# 39. Skill 7 — Basic data visualization

Start with:

- matplotlib

- Plotly

Plotly will become particularly useful because it supports interactive visualizations.

Learn:

- scatter plots
- labels
- tooltips
- animation frames
- sliders

Your first gravity prototype can simply be a Plotly scatter plot.

No planets yet.

Just circles.

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# 40. Skill 8 — Streamlit

Streamlit lets you transform Python analysis into a web application without becoming a full web developer.

You can create:

- dropdowns
- sliders
- buttons
- graphs
- text panels

using Python.

Perfect for this project.

You do not initially need:

- React
- JavaScript
- complex frontend frameworks

Build it with Streamlit first.

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# 41. Skill 9 — Git and GitHub

This matters because this is a portfolio project.

Learn:

- repository
- commit
- push
- pull
- branches eventually
- README

You want HCSS to see:

> This person can build something reproducibly.

Not:

> Here is a random Jupyter notebook.

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# 42. Skill 10 — Basic SQL

This can come slightly later.

As your dataset grows, store it in:

**SQLite**

You can then learn simple SQL:

SELECT
WHERE
GROUP BY
JOIN

Later:

PostgreSQL.

For the MVP, CSV/Parquet files are completely acceptable.

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# 43. Skill 11 — Topic modelling

Only learn this after your embedding pipeline works.

Start with:

**BERTopic**

Learn conceptually:

embeddings
→ clustering
→ recurring words
→ topics

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# 44. Skill 12 — Validation

This one matters enormously.

Do not blindly trust the model.

Take a sample, perhaps:

**200 paragraphs**

Then manually label them yourself.

For example:

Ukraine relevance:

- yes/no

Sanctions stance:

- support
- oppose
- neutral

Negotiation framing:

- high
- low

Then compare your NLP system with your own coding.

This is called:

## Ground truth

Your manually coded sample becomes the approximate answer against which the model is evaluated.

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# 45. Evaluation metrics

Eventually learn:

### Accuracy

How often was the model correct?

### Precision

When the model says something belongs to a category, how often is it actually correct?

### Recall

Of all the relevant cases, how many did the model find?

### F1 score

A combination of precision and recall.

You do not have to memorize formulas initially.

Understand the questions they answer.

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# 46. Avoid LLM magic

You could technically send every speech to an LLM and ask:

> “Rate this country from Russia to Ukraine 1–100.”

Do **not** make that your project.

It would be:

- difficult to validate
- difficult to reproduce
- difficult to explain
- vulnerable to prompt changes
- methodologically weak

LLMs can eventually help with:

- structured extraction
- stance coding
- summarization
- naming topics

But your central scores should have a transparent methodology.

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# 47. LLMs should support the pipeline, not replace the analysis

A good architecture might eventually be:

Official speech
↓
Paragraph extraction
↓
Issue relevance
↓
Embedding
↓
Stance extraction
↓
Narrative dimensions
↓
Statistical aggregation
↓
Gravity score
↓
Visualization
↓
Evidence

An LLM can assist in some boxes.

It should not simply replace all of them.

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# 48. Your application structure

## Page 1 — Global Narrative Universe

Main visual.

Controls:

**Issue**
Russia–Ukraine ▼

**Year**
2024

**Show trails**
Yes

**Region**
All

Large interactive narrative space.

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# 49. Page 2 — Country Explorer

Select:

🇮🇳 India

Show:

### Current narrative profile

International law [redacted]
Negotiation [redacted]
Military support [redacted]
Sanctions [redacted]
Multipolarity [redacted]

Then:

### Narrative trajectory

2014–2025

Then:

### Closest states

Brazil
South Africa
Indonesia

Then:

### Evidence

Actual paragraphs.

---

# 50. Page 3 — Compare Countries

Choose:

India

versus

Poland

Display:

- narrative positions
- topics
- stance differences
- trajectory
- representative statements

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# 51. Page 4 — Narrative Shifts

Automatically identify:

## Largest changes this year

1. Germany — 2022
2. Finland — 2022

3. Sweden — 2022
4. Hungary — ...
5. Türkiye — ...

Clicking an observation explains the change.

That feels very much like an analyst's tool.

---

# 52. Page 5 — Narrative versus Behaviour

Later version.

Plot:

X-axis:

Narrative proximity

Y-axis:

UN voting proximity

Countries in unusual positions become analytically interesting.

---

# 53. Very important: uncertainty

Your system should display how much evidence exists.

Example:

**Narrative position confidence: LOW**

Reason:

Only one relevant paragraph identified.

Versus:

**Confidence: HIGH**

14 relevant paragraphs identified with consistent framing.

This will make the project considerably more mature.

---

# 54. What your first successful prototype needs

Your first version does NOT need:

- every country
- every issue
- an LLM
- real-time scraping
- sophisticated databases
- beautiful 3D graphics
- prediction
- neural network training

It needs only:

1. UN speeches
2. approximately 10 countries
3. several years
4. one issue
5. paragraph extraction
6. embeddings
7. similarity scores
8. simple visualization
9. evidence retrieval

If that works, your central idea is proven.

---

# 55. Recommended development roadmap

## WEEK 1 — Understand the data

Goal:

Get speeches into Python.

Tasks:

Download the UN General Debate Corpus.

Load it into Pandas.

Understand:

- country
- year
- text

Choose:

- Russia
- Ukraine
- US
- Germany
- France
- India
- Brazil
- South Africa
- China
- Türkiye

Use years:

2021–2025.

Do nothing fancy.

At the end of Week 1 you should be able to write:

```
speech = data[
    (data["country"] == "India") &
    (data["year"] == 2023)
]
```

and obtain India's speech.

That alone is progress.

---

# 56. WEEK 2 — Break speeches into paragraphs

Goal:

Create your first text-analysis dataset.

One row:

country
year
paragraph

Build perhaps several thousand paragraphs.

Then manually inspect them.

Search terms:

Ukraine
Russia
war
invasion
territorial
NATO
sanctions
peace
ceasefire

For the first prototype, keyword filtering is acceptable.

Do not worry that it is basic.

You are establishing the pipeline.

---

# 57. WEEK 3 — Learn embeddings

Goal:

Turn paragraphs into embeddings.

Install:

sentence-transformers

Use a pretrained model.

Encode paragraphs.

Then test questions yourself.

Take:

Russia must withdraw from Ukraine.

Find the ten most similar paragraphs in the corpus.

Read them.

Do they make sense?

This teaches you embeddings far better than watching ten hours of videos.

---

# 58. WEEK 4 — Build your first narrative comparison

Create:

Russian anchor texts.

Ukrainian anchor texts.

Calculate the similarity of other countries' relevant paragraphs to each.

Aggregate them by:

country × year.

Then create a very basic chart.

Something as simple as:

Ukraine ←----------→ Russia

Poland ●

Germany ●

India ●

Brazil ●

Do not build the galaxy yet.

Make sure the numbers make political sense first.

---

# 59. WEEK 5 — Add stance and dimensions

Now improve the system.

Create your narrative dimensions:

- territorial integrity
- sanctions
- negotiation
- NATO
- responsibility
- escalation

Manually code examples.

Experiment with:

- zero-shot classification
- embedding similarity
- possibly an LLM for comparison

Compare results.

This is where the actual methodological research begins.

---

# 60. WEEK 6 — Validate it

Select perhaps 150–300 paragraphs.

Code them manually.

Compare:

your judgement
versus
model prediction.

Calculate basic evaluation measures.

Document the mistakes.

This is impressive because most portfolio projects skip this entirely.

---

# 61. WEEK 7 — Build Strategic Gravity

Now you have permission to make it beautiful.

Use:

Plotly + Streamlit.

Create:

- narrative poles
- country points
- year slider
- trajectories
- tooltips
- clickable country panels

Focus on readability rather than creating literal 3D planets.

---

# 62. WEEK 8 — Turn it into a policy product

Polish:

### GitHub repository

### interactive application

### methodology page

### data documentation

### limitations

### 2–4 page analytical brief

Potential brief:

> **From Condemnation to Negotiation: Mapping Global Strategic Narratives on the Russia–Ukraine War, 2021–2025**

This brief could highlight 3–5 interesting countries.

For example:

- India
- Brazil
- Hungary
- Türkiye
- South Africa

Now the project becomes something you can send to HCSS.

---

# 63. GitHub structure

Eventually something like:

```
global-strategic-narratives/
│
├── README.md
│
├── requirements.txt
│
├── data/
│   ├── raw/
│   └── processed/
│
├── notebooks/
│   ├── 01_explore_data.ipynb
│   ├── 02_topic_filtering.ipynb
│   ├── 03_embeddings.ipynb
│   └── 04_validation.ipynb
│
├── src/
│   ├── preprocessing.py
│   ├── embeddings.py
│   ├── similarity.py
│   ├── stance.py
│   └── visualization.py
│
├── app/
│   └── streamlit_app.py
│
├── methodology/
│   └── methodology.md
│
└── outputs/
```

You will not understand all of this immediately.

That is fine.

You gradually move code out of notebooks into reusable Python files.

---

# 64. What I would use technically

## Main language

### Python

You can use R later for statistical analysis if you prefer.

But Python is better for this particular project because of the NLP ecosystem.

## Core tools

**pandas**
data manipulation

**sentence-transformers**
embeddings

**scikit-learn**
similarity, clustering, modelling

**spaCy**
text processing

**Plotly**
interactive visualizations

**Streamlit**
web application

**BERTopic**
later topic modelling

**Git / GitHub**
version control and public portfolio

**SQLite/PostgreSQL**
later database

---

# 65. What you do NOT need to learn now

Do not disappear down these rabbit holes yet:

- advanced deep-learning mathematics
- training transformers from scratch
- PyTorch architecture
- React
- Kubernetes
- cloud infrastructure
- graph neural networks
- agentic AI systems
- complex MLOps

All of those can come later.

Your objective is to understand the analytical pipeline.

---

# 66. Validation beyond statistics

Your international-relations training should become part of the validation.

For every surprising result ask:

> Does this make substantive political sense?

If India suddenly jumps beside Ukraine in 2023, investigate.

Maybe:

- the model is wrong
- the relevant paragraph is unusual
- the government genuinely changed rhetoric
- there is too little text
- the speech discusses Ukraine only to criticize something
- translation creates problems

This interaction between:

**model result**

and

**political interpretation**

is exactly what makes the project interesting.

---

# 67. Important limitation: rhetoric is not foreign policy

Write this clearly everywhere.

Your tool measures:

**publicly expressed strategic discourse.**

It does not directly measure:

- secret policy
- diplomatic intentions
- military alignment
- public opinion
- elite beliefs
- actual behaviour

This isn't a weakness if you state it correctly.

It is the phenomenon you chose to study.

---

# 68. Important limitation: UN speeches are strategic performances

Governments know they are speaking publicly.

Therefore these speeches are not raw windows into what governments “really think.”

They are themselves political acts.

That is actually interesting.

Your interpretation becomes:

> We measure how governments choose to publicly frame international issues.

Not:

> We discover their true beliefs.

---

# 69. Important limitation: narrative poles are researcher choices

If you decide that:

Russia

Ukraine

US/NATO

are the important poles, you are making a theoretical decision.

You need to explain and justify it.

Eventually you can compare:

**researcher-defined anchors**

versus

**data-driven clusters**.

If both produce similar structures, that strengthens your findings.

---

# 70. Important limitation: language

Fortunately, starting from standardized English UN transcripts reduces this problem.

Once you scrape national sources, language becomes much harder.

You will then need to think about:

- multilingual embeddings
- translation
- cross-language comparability

That is a Phase 2 problem.

---

# 71. The global version

Once your Ukraine prototype works, the final product becomes:

# Strategic Gravity

User selects:

### ISSUE

Russia–Ukraine

### YEAR

2024

Then sees the narrative universe.

Switch:

### ISSUE

US–China competition

The universe reorganizes.

Switch:

### ISSUE

Global multipolarity

Again the positions change.

That is the important idea:

> There is not one permanent geopolitical map.

There are multiple overlapping strategic narrative spaces.

---

# 72. The really powerful visual insight

Imagine clicking:

🇮🇳 India.

Instead of one position, show:

### India across strategic spaces

Russia–Ukraine
→ negotiation / non-alignment

US–China
→ moderate / strategic autonomy

Multipolarity
→ relatively close to Global South/BRICS framing

International institutions
→ strong multilateralist framing

Now your system begins representing the complexity of international relations much better than a simple West-versus-East scale.

---

# 73. Network extension

Eventually another visualization could represent countries as a network.

Countries with similar narratives become connected.

For example:

India ——— Brazil
| |
South Africa ─ Indonesia

Another cluster:

Poland ─ Netherlands ─ UK

You can then study:

## Narrative communities

These might overlap with alliances.

Or they might not.

The differences are analytically interesting.

---

# 74. Prediction should come much later

Do not initially call this an:

**early-warning system.**

First prove that you can reliably measure narrative movement.

Later you could investigate:

> Do unusually large narrative changes precede changes in geopolitical behaviour?

For example:

Narrative movement at time (t)

predicting:

voting change at (t+1)

or:

diplomatic events at (t+1)

That could become a statistical thesis-level extension.

---

# 75. What makes this different from a normal portfolio project

A normal NLP project says:

> I classified sentiment in political speeches.

Your project says:

> I designed a measurable conceptual framework for strategic narratives, created a longitudinal global political-text dataset, tested different NLP approaches, validated them manually, developed indicators of narrative proximity and change, and turned the results into an interactive policy-analysis tool.

That is a completely different level.

---

# 76. What the project demonstrates professionally

It demonstrates:

### International relations

You understand state behaviour and strategic narratives.

### Political science

You understand concepts, operationalization and measurement.

### Statistics

You understand uncertainty, validation and longitudinal analysis.

### NLP

You understand embeddings, classification, clustering and retrieval.

### Data engineering

You can collect and structure textual data.

### Visualization

You can transform complicated results into something intuitive.

### Policy communication

You can translate methods into strategic insight.

That combination is the entire value proposition of the project.

---

# 77. Why it fits HCSS

The conceptual match with HCSS is particularly strong.

Their Datalab works at the intersection of:

- defence
- security
- geopolitics
- quantitative methods
- NLP
- AI
- horizon scanning
- interactive analytical products

Your project should therefore deliberately demonstrate:

**data + methodology + geopolitical understanding + usable visual product.**

However, do not try to replicate an HCSS platform.

Your unique research proposition should remain:

> **Measuring and explaining how states' strategic narratives themselves move through time and across issues.**

---

# 78. Final deliverables

When finished, you should have five things.

## 1. Interactive web application

Strategic Gravity.

## 2. GitHub repository

Clean and reproducible.

## 3. Methodology note

Perhaps 5–10 pages explaining:

- data
- concepts
- NLP
- scoring
- validation
- limitations

## 4. Short policy brief

Approximately 3 pages.

Not technical.

It shows actual geopolitical findings.

## 5. One-minute demo video/GIF

Show:

2021 → 2022 → 2023 → 2024 → 2025

and watch countries move.

This will be excellent for LinkedIn and for contacting researchers.

---

# 79. Your learning philosophy

Do not try to “learn NLP” first and build the project afterward.

Build the project while learning.

The sequence should be:

**I need to load data**

→ learn Pandas.

**I need to understand meaning**

→ learn embeddings.

**I need to compare meaning**

→ learn cosine similarity.

**The model confuses discussing Russia with supporting Russia**

→ learn stance detection.

**I need to find themes**

→ learn topic modelling.

**I need to know whether the model works**

→ learn validation.

**I need users to interact with it**

→ learn Streamlit.

This way every technical skill solves a problem you actually understand.

---

# 80. Your first three tasks

Do not begin with the galaxy visualization.

Start here.

### Task 1

Download the UN General Debate Corpus.

Load it into Python.

Explore the country/year/text variables.

### Task 2

Create a subset containing:

Russia
Ukraine
US
China
India
Brazil
South Africa
Germany
France
Türkiye

for:

2021–2025.

Read several speeches manually.

### Task 3

Split those speeches into paragraphs and create:

country
year
paragraph_id
paragraph_text

Once you have that table, your actual NLP project has begun.

---

# 81. The long-term vision

The eventual system could answer questions like:

> Which states have shifted most strongly toward negotiation-based rhetoric on Ukraine?
>
> Is Global South rhetoric on the war converging or becoming more fragmented?
>
> Which countries rhetorically resemble China on multipolarity but vote more closely with Western states?
>
> Has European discourse become more internally homogeneous since 2022?
>
> Which states changed their strategic framing most dramatically after a geopolitical shock?
>
> Are there emerging narrative coalitions that do not correspond to formal alliances?

Those are no longer toy NLP questions.

They are real international-relations questions that data science can help answer.

---

# 82. Final definition of the project

**The Global Strategic Narrative Monitor is an interactive computational political-analysis platform that uses NLP to measure, visualize and explain changes in states' official strategic narratives across geopolitical issues and over time. Its Strategic Gravity interface represents narrative proximity as movement between issue-specific geopolitical narrative poles, while an evidence layer allows every position and movement to be traced back to official source text. The project combines quantitative text analysis with political-science interpretation and eventually compares rhetorical positioning with observable state behaviour.**

That is the project I would build.